# Introduction to Operation Management

## PART TWO Design the operation

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# 1 Process design

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## What is process design

Design is indeed a multifaceted activity that encompasses both the conceptual and practical aspects of creating something new or improving an existing entity. It involves envisioning and planning before actual implementation, and it requires a balance between creativity and functionality.

In the context of process design, the approach is systematic and iterative. Initially, the focus is on understanding the broader objectives and constraints. This high-level view is crucial for setting the direction and scope of the design. The process is often characterized by its volume (how much output it needs to produce) and variety (the range of outputs or the flexibility required). These characteristics help in determining the type of process that is suitable, such as a mass production line for high volume and low variety, or a more flexible setup for low volume and high variety.

As the design progresses, more detailed considerations come into play. This involves delving into the specifics of how the process will operate, including the sequence of operations, the technology and equipment needed, the layout of the workspace, and the skills required by the workforce. Each detail is critical to ensuring that the process is efficient, cost-effective, and capable of meeting quality standards.

Throughout the design process, there is a constant evaluation of how well the design meets the initial objectives and whether it is feasible in a real-world context. This may involve creating prototypes, simulations, or models to test and refine the design before full-scale implementation.

Design is a layered activity that starts with a broad vision and progressively narrows down to specific details, all while maintaining a clear focus on the end goals and practical viability.

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## Process design and product/service design are interrelated

The interdependence between the design of products and services and the design of the processes that create them is a critical aspect of operations management. This relationship is often referred to as the "design-process nexus," and it highlights the importance of considering both elements as part of a cohesive system.

In product and service design, functionality, aesthetics, user experience, and cost are primary considerations. However, these designs must also take into account the capabilities and limitations of the production processes. For instance, a product designed with complex features might require sophisticated manufacturing techniques, specialized materials, or additional labor, which could increase production costs and complexity. Conversely, a process-oriented design approach might prioritize ease of manufacture, assembly, and serviceability, potentially influencing the final product's features and performance.

The design of services is particularly intertwined with process design because the customer often directly experiences the process. In service operations, the process is not just a means to an end but is part of the service itself. For example, the layout of a restaurant or the interface of a banking app directly affects the customer's experience. Therefore, service design must consider the customer's journey through the process, ensuring that it is efficient, pleasant, and aligned with the service's intended value proposition.

The historical example of aircraft engineers also being test pilots illustrates the benefits of designers having a deep understanding of the production and use of their designs. When designers are involved in the making or using of their products, they gain firsthand knowledge of practical challenges, user needs, and the implications of design choices. This can lead to more informed, practical, and user-centered designs.

In modern operations, the integration of product and process design is facilitated by various methodologies and tools, such as concurrent engineering, design for manufacture and assembly (DFMA), and agile development. These approaches encourage cross-functional collaboration, early consideration of production issues, and iterative design processes that incorporate feedback from both the production side and the end-users.

Overall, recognizing and managing the interrelationship between product/service design and process design is essential for creating successful operations that deliver value to customers while maintaining efficiency and competitiveness.

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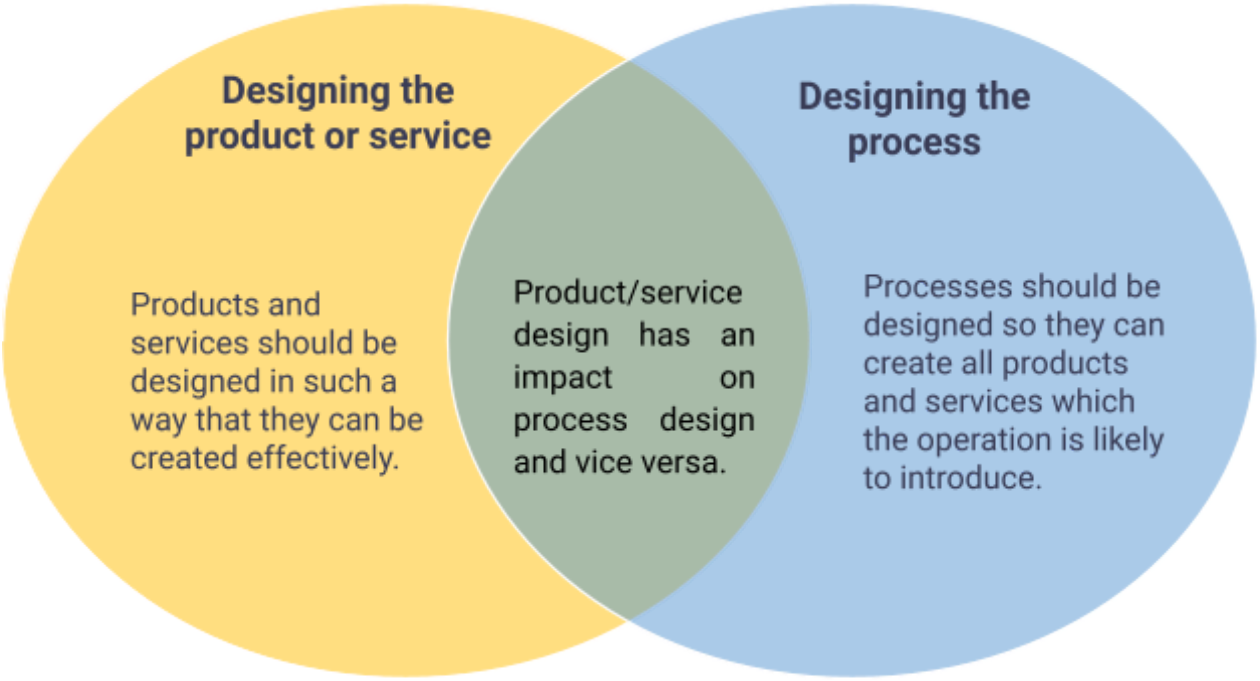
FIGURE 1. The design of products/services and the design of processes are interrelated and should be treated together.

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## References

Slack, N. Brandon-Jones. (2019). *Operations Management*. Pearson

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## Network planning

Network planning is a strategic approach to managing and optimizing a firm's supply chain. It involves a series of decisions and actions that aim to align the supply chain's capabilities with the company's business objectives. The primary goals of network planning include:

1. Balancing Costs: Network planning seeks to find the optimal balance between various supply chain costs, such as inventory holding costs, transportation expenses, and manufacturing costs. By analyzing these costs in relation to each other, a company can determine the most cost-effective way to structure its supply chain.
2. Matching Supply and Demand: Effective network planning ensures that the supply of products meets customer demand without excessive overproduction or stockouts. This involves forecasting demand accurately, setting appropriate inventory levels, and planning for capacity to respond to demand fluctuations.
3. Managing Inventory: Inventory management is a critical component of network planning. It involves determining the right amount of inventory to hold at different points in the supply chain to meet service level targets while minimizing costs. This includes decisions about safety stock, reorder points, and economic order quantities.
4. Sourcing from Optimal Locations: Network planning involves deciding where products should be manufactured and how they should be distributed. This includes selecting the most appropriate manufacturing facilities based on factors such as production costs, capacity, proximity to markets, and the availability of raw materials.

To achieve these goals, network planning may involve various analytical and optimization techniques, such as:

- Supply Chain Modeling: Creating models of the supply chain to simulate different scenarios and assess the impact of various decisions on costs and service levels.
- Optimization Algorithms: Using mathematical algorithms to find the best solutions for complex problems, such as facility location, transportation routing, and inventory policies.
- Risk Analysis: Evaluating potential risks in the supply chain, such as supplier reliability, demand variability, and geopolitical factors, and developing strategies to mitigate these risks.
- Collaborative Planning: Working closely with suppliers, manufacturers, and logistics providers to coordinate activities and share information for better decision-making.

Network planning is an ongoing process that requires continuous monitoring and adjustment as market conditions, customer preferences, and other external factors change. By effectively managing their supply chain networks, firms can improve their competitiveness, enhance customer satisfaction, and achieve greater operational efficiency.

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The network planning process can indeed be divided into several key steps, each focusing on different aspects of the supply chain.

1. Network Design

- Number of Facilities: Deciding on the number of manufacturing plants and warehouses required to meet the company's production and distribution needs.

- Locations: Determining the geographical placement of these facilities to optimize proximity to suppliers, customers, and transportation hubs.

- Size of Facilities: Establishing the scale of each facility, including its capacity for production, storage, and throughput, to handle current and projected demand.

2. Inventory Positioning:

- Identifying Stocking Points: Deciding where in the supply chain to hold inventory, which could include raw materials at manufacturing sites, finished goods at warehouses, or products at distribution centers.

- Facilities Producing to Stock (Make-to-Stock): Selecting facilities that will maintain an inventory of finished goods to meet anticipated customer demand based on forecasts. This approach is common for products with predictable demand patterns.

- Facilities Producing to Order (Make-to-Order): Identifying facilities that will produce goods only after receiving a customer order, thereby keeping little to no inventory of finished products. This approach reduces inventory holding costs and is often used for customized or low-volume products.

3. Resource Allocation:

- Production Allocation: Determining which products should be produced at which facilities based on factors such as manufacturing capabilities, cost efficiencies, and capacity.

- Packaging Allocation: Deciding where the packaging of different products will take place, which can be at the manufacturing site or at a separate facility closer to the point of distribution.

- Right Facility: Ensuring that each facility is utilized for its intended purpose and that it aligns with the overall network strategy. This includes assessing whether a facility is best suited for high-volume production, specialized manufacturing, quick-turnaround orders, or other specific roles within the supply chain.

Each of these steps involves complex decision-making and may require the use of advanced analytical tools, such as optimization software, simulation models, and data analytics. The goal is to create a supply chain network that is resilient, responsive, and aligned with the company's service level expectations and cost objectives.

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## Network design

Thinking about operations as networks of processes and as part of larger networks is crucial for achieving operational excellence.It has many processes that transform items and transfer them to other internal processes. Through this network there are many 'process chains', that is, threads of processes within the network. And thinking about processes as part of a network has several advantages. It helps to align individual processes with the strategic objectives of the organization, ensures that employees understand the value of their work in the context of the customer experience, and promotes a collaborative environment where the interconnections between processes are managed effectively.

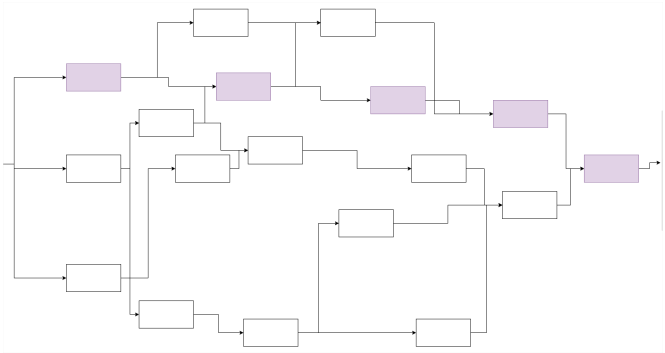

FIGURE 2. A process network within an operation showing an internal process chain

### Objectives of a process design

The objective of process design in the context of operations as networks should be to create a system that is efficient, effective, and aligned with the strategic goals of the organization.

#### Quality

Incorporate quality control measures within the process to ensure that the output is defect-free and adheres to established standards.

Implement continuous improvement practices to enhance quality over time.

#### Efficiency:

Maximize the use of resources, minimize waste, and reduce costs without compromising quality or performance.

Streamline workflows to eliminate unnecessary steps and reduce cycle times.

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### Effectiveness:

Design processes that consistently produce outputs that meet or exceed customer requirements and expectations.

Ensure that the process delivers value to the customer and the organization.

### Flexibility and Scalability:

Allow for adjustments to be made to the process in response to changing market conditions, customer demands, or technological advancements.

Design processes that can be scaled up or down as needed without significant reengineering.

### Compliance and Sustainability:

Ensure that processes comply with legal, regulatory, and ethical standards.

Design processes that are environmentally sustainable and socially responsible.

## Volume and variety in process design

Volume and variety are two critical factors that significantly influence process design in operations management. They are part of the four V's (volume, variety, variation in demand, and visibility) that characterize the nature of demand for goods or services. Here's how volume and variety affect process design:

1. Volume:

- High Volume: High-volume processes typically benefit from standardization and automation. When the volume of production is large, it is cost-effective to invest in specialized equipment and technology that can produce goods or deliver services quickly and consistently. High volume allows for economies of scale, which can reduce the unit cost of production. Process design for high-volume operations often focuses on efficiency, throughput, and minimizing unit costs.

- Low Volume: Low-volume processes may require more flexibility and may not justify the same level of investment in automation. In these cases, process design might emphasize the ability to handle a wider range of tasks or to switch between different products or services quickly. Low-volume operations might use more general-purpose equipment and rely more on skilled labor.

2. Variety:

- High Variety: When there is a high variety of products or services, the process design needs to accommodate customization and changeovers. Processes must be flexible to handle different product specifications, customer requirements, or service needs. High variety can lead to complexity in scheduling, inventory management, and quality control. Process design for high-variety operations often includes modular or cellular layouts, flexible workforce, and quick setup procedures.

- Low Variety: Low variety allows for more repetition and standardization in the process. With a limited range of products or services, it is easier to optimize the process for efficiency. Process design can focus on repetitive tasks, which can be

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refined and improved over time. Low variety often leads to a more predictable process flow and can simplify inventory management and quality assurance.

The relationship between volume and variety can be further understood through the product-process matrix, which suggests that certain process types are more suitable for specific combinations of volume and variety:

- Job Shop: Low volume, high variety. Customized products or services with a flexible process.

- Batch Production: Medium volume, medium variety. A series of products or services produced in sets or batches.

- Mass Production: High volume, low variety. Standardized products with a continuous or assembly line process.

- Continuous Production: Very high volume, very low variety. Highly automated and continuous flow processes.

Process design must take into account the trade-offs between volume and variety to ensure that the operation can meet demand efficiently and effectively. Here's how these trade-offs continue to influence process design:

- Customization vs. Standardization: High variety often requires customization, which can conflict with the desire for the standardization associated with high volume. Process design must find a balance between these two extremes, potentially through the use of flexible manufacturing systems or by offering a standardized base product with customizable options.

- Lead Time: High volume operations can achieve shorter lead times through economies of scale, while high variety may increase lead times due to the need for changeovers or customization. Process design should aim to minimize lead times within the constraints of volume and variety.

- Cost Implications: High volume typically leads to lower costs per unit due to economies of scale, but high variety can increase costs due to the need for more complex processes, additional inventory, and potentially lower utilization of equipment. Process design should consider cost-effective ways to manage variety without sacrificing the benefits of high volume.

- Workforce Skills: High-variety processes may require a more skilled and flexible workforce capable of handling different tasks and adapting to changes. In contrast, high-volume processes may benefit from a more specialized workforce focused on efficiency and consistency. Process design should align with workforce capabilities and training needs.

- Equipment and Technology: The choice of equipment and technology is also influenced by volume and variety. High-volume operations might use dedicated machinery, while high-variety operations might opt for more versatile equipment. Process design should consider the appropriate level of automation and technology investment.

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- Layout and Flow: The physical layout of the operation must support the process design. High-volume operations might use a product layout optimized for a smooth flow of materials, while high-variety operations might use a process layout that groups similar activities together to handle different products or services.

- Supply Chain Integration: Volume and variety affect not only the internal processes but also the broader supply chain. High-volume operations may work with suppliers to ensure a steady flow of materials, while high-variety operations may need suppliers that can provide a wider range of components or materials quickly and flexibly.

In summary, process design must carefully consider the implications of volume and variety to create an operation that is responsive to customer needs while maintaining efficiency and controlling costs. The goal is to design processes that are robust enough to handle the expected range of volume and variety while remaining agile enough to adapt to changes

## How do are process designed in detail BPMN

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## References

Simchi-Levi, D. (2011), *Designing and Managing the Supply Chain*. Mc Graw Hill

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